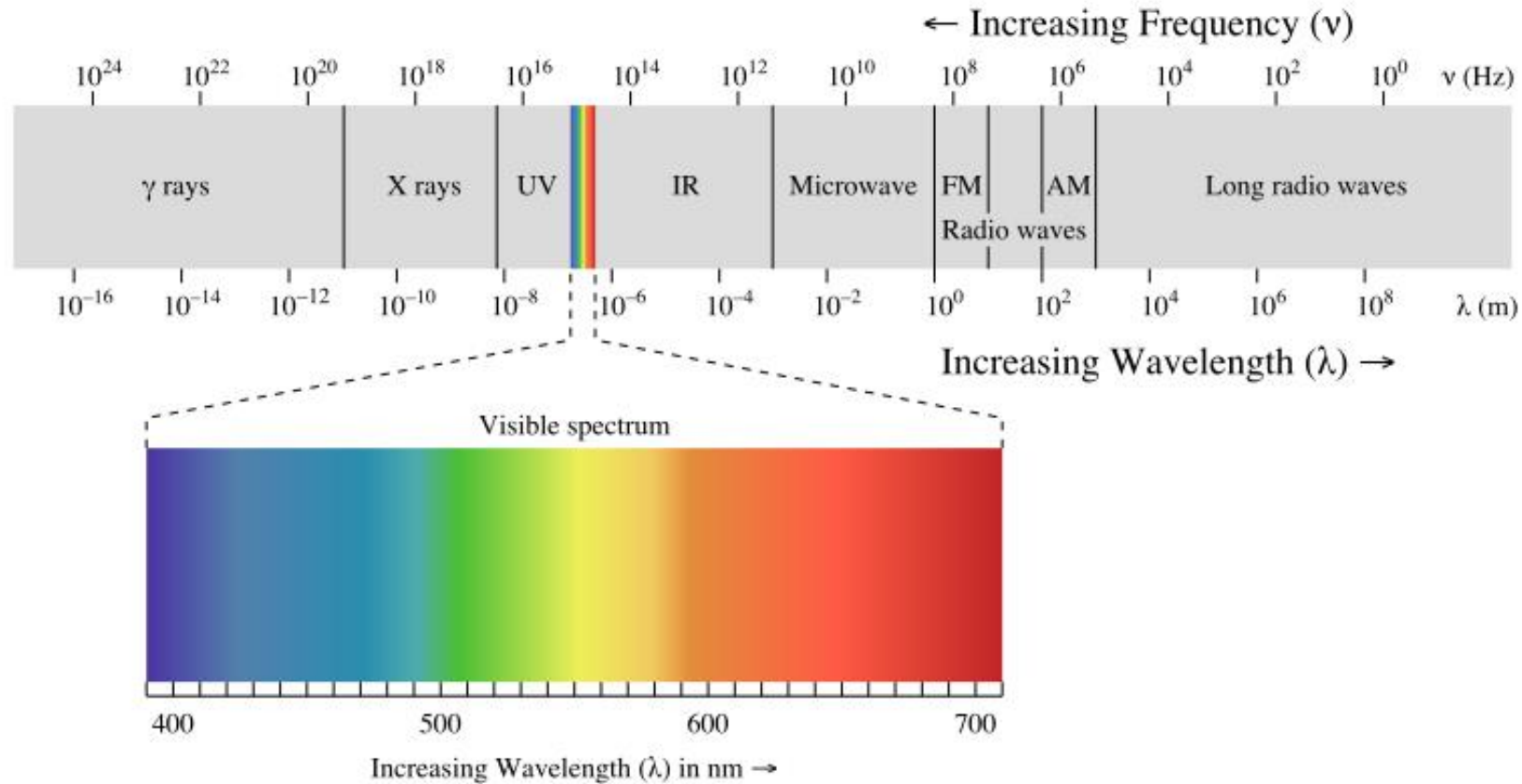


OPTICAL FIBER

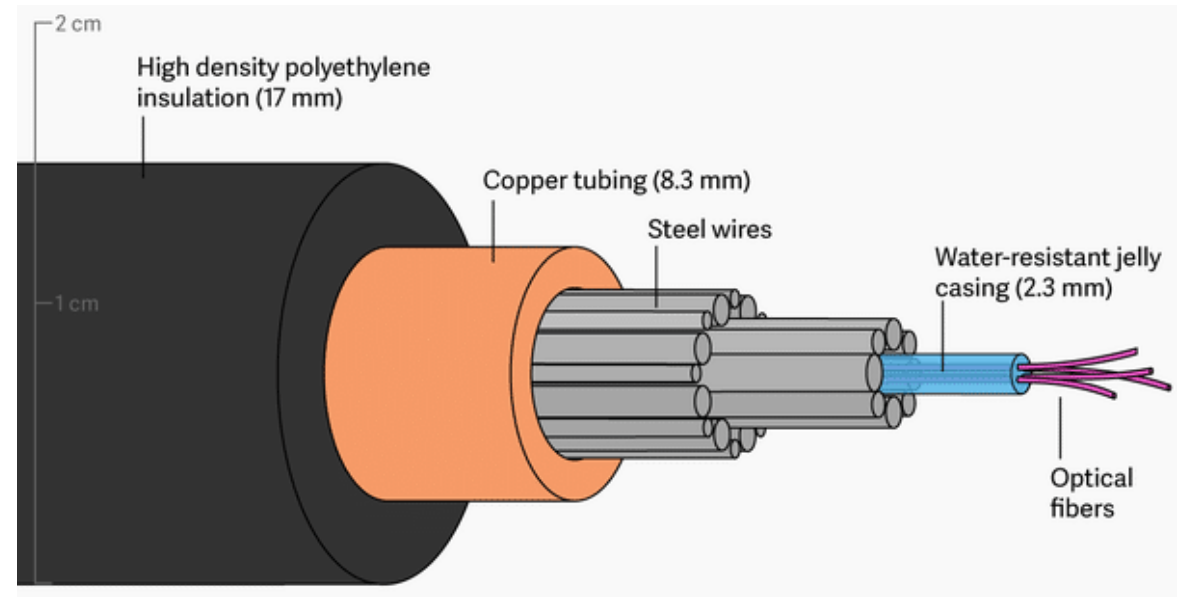
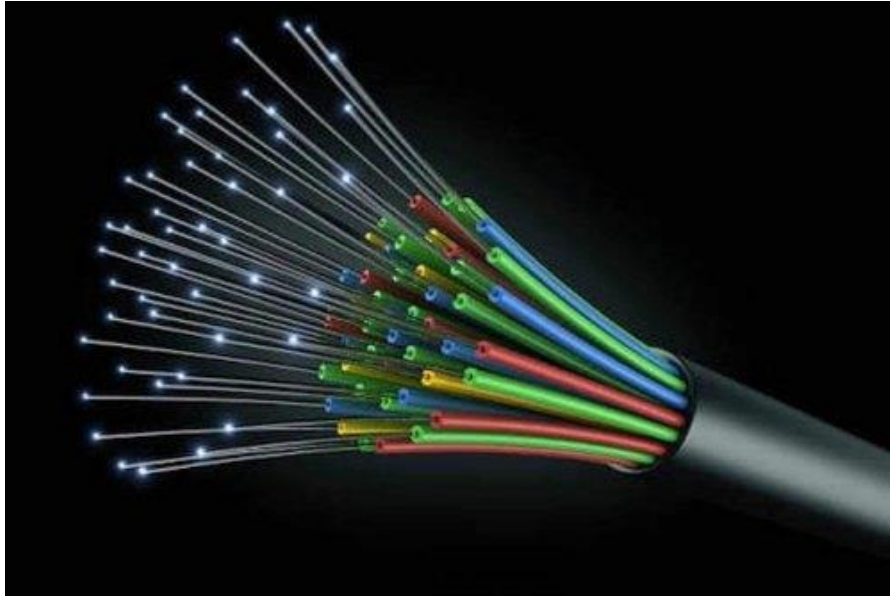
Electromagnetic Waves : Data Communications



Radio waves (MHz) & Microwaves (GHz) are generally used for data communication system. Optical fiber communication uses Light waves (visible spectrum).

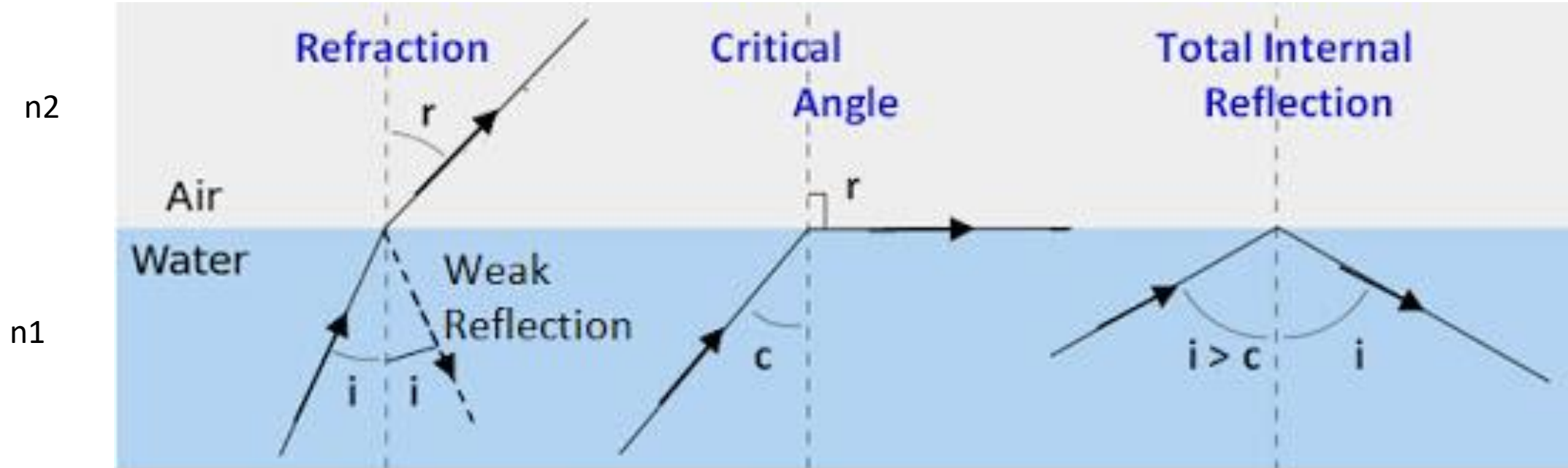
An **optical fiber** is a thin **fiber** of glass or plastic with diameter comparable to a human hair that can carry light from one end to the other. The study of **optical fibers** is called **fiber optics**.

They are used in fiber optic communication for transmission of data to much longer distances.



The propagation of light through the optical fiber cable is based on the phenomenon of total internal reflection.

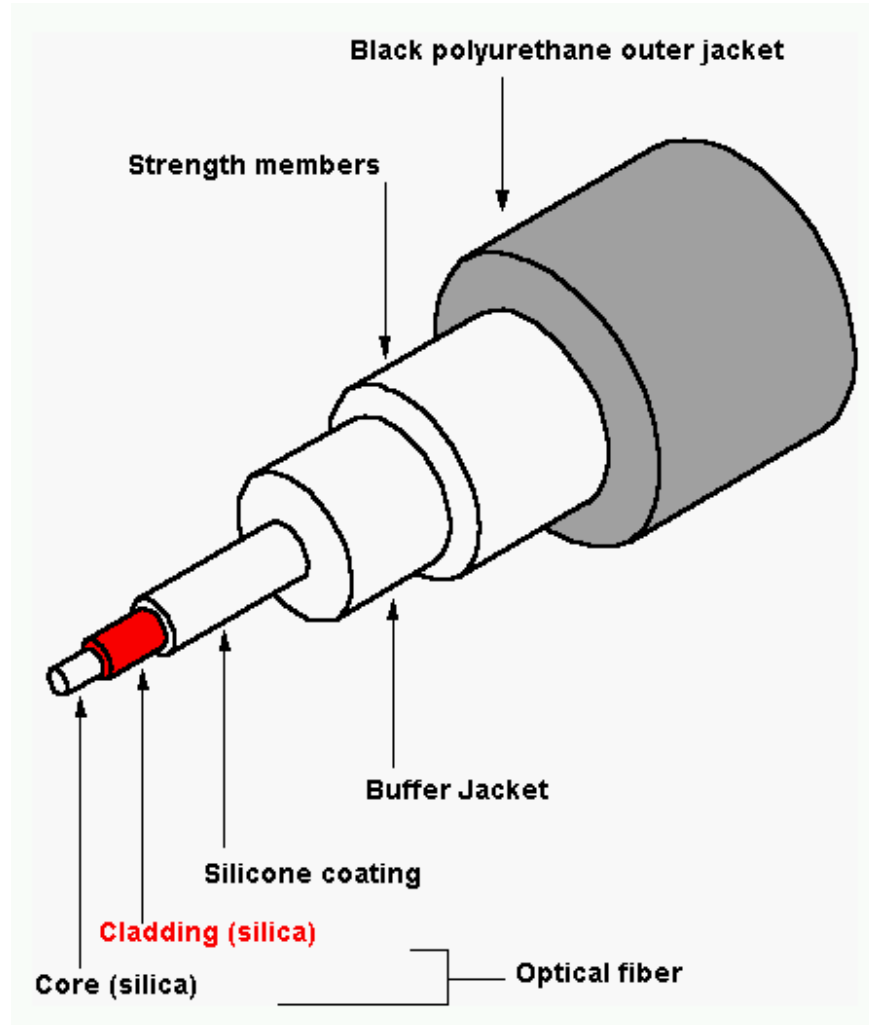
Total Internal Reflection (TIR)



Snell's Law
 $n_1 \sin i = n_2 \sin r$

Snell's Law
 $n_1 \sin c = n_2 \sin 90$
 $\sin c = n_2/n_1$

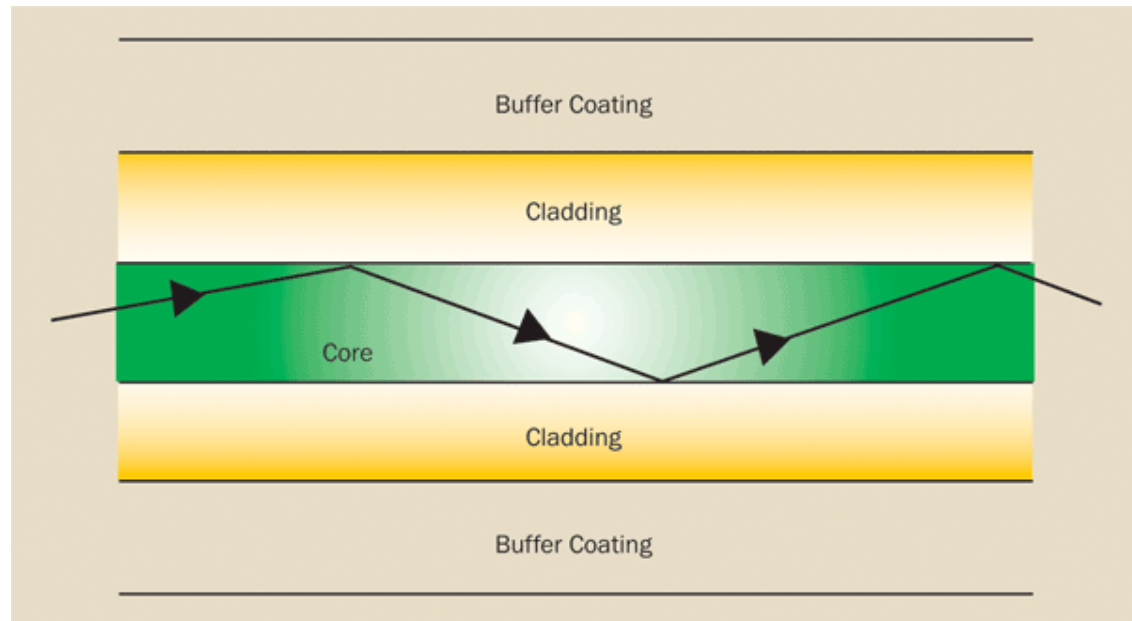
Structure of optical fiber



Important parts of optical fiber are core and cladding.

Only core and cladding participate for propagation of light. Light wave propagate inside the core region.

Refractive index of core is more than the refractive index of cladding
 $n(\text{core}) > n(\text{cladding})$



Types of optical fibers

Transverse cross sections of different types of optical fibers

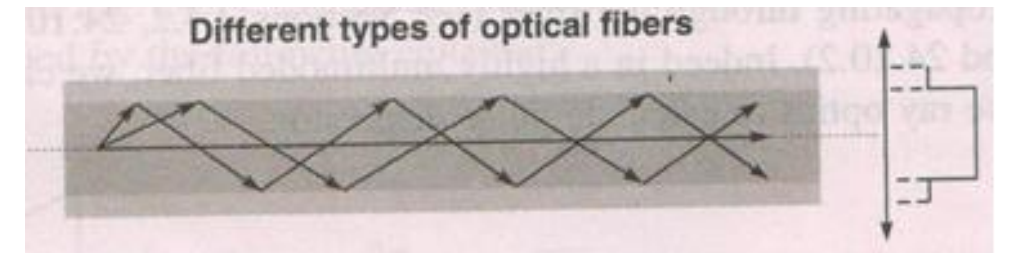
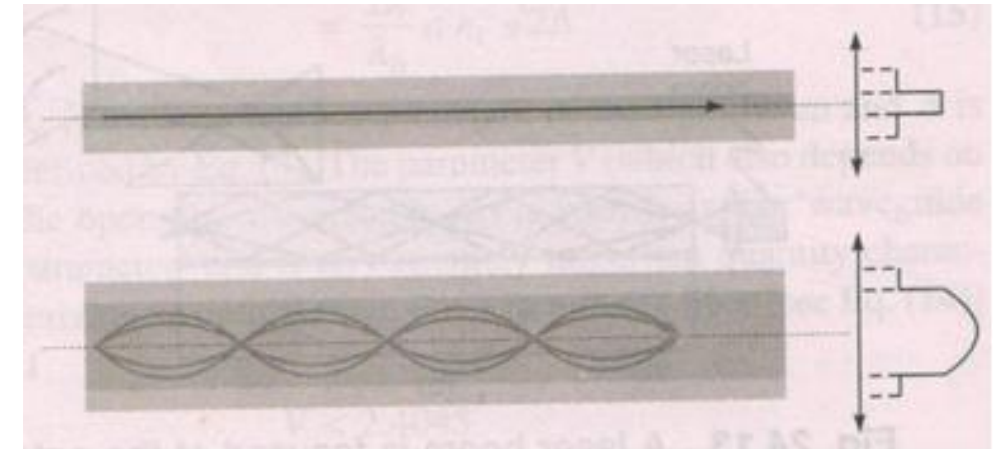
**Single mode
step index
fiber**



**Graded
index fiber**

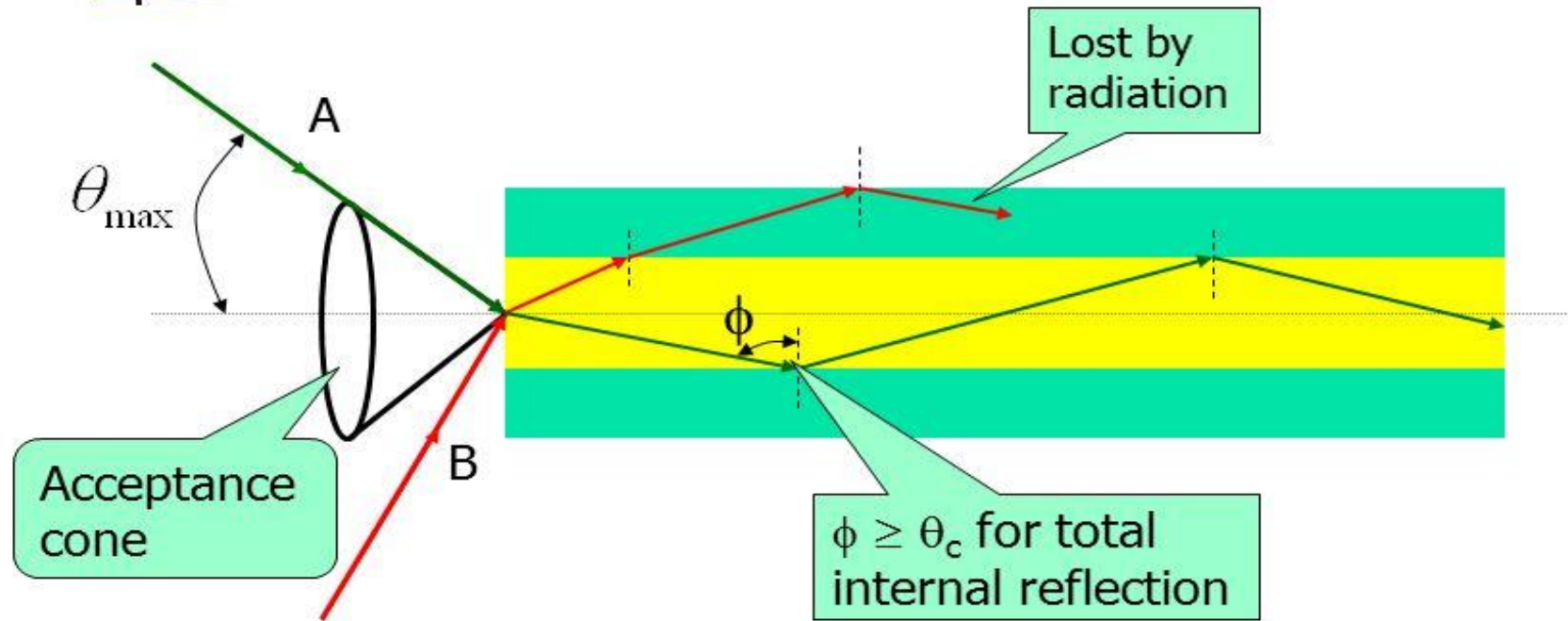


**Multi mode
step index
fiber**



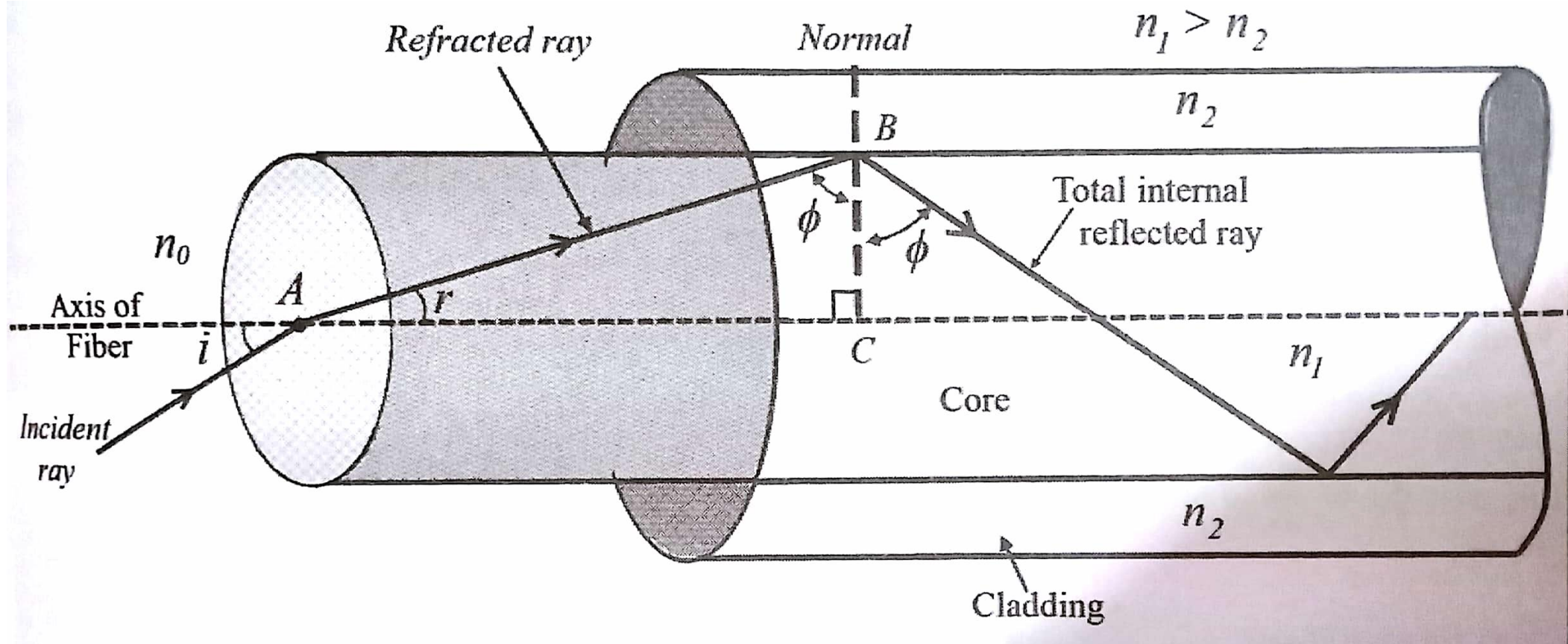
$$\text{Numerical Aperture} = \sin i_m = \sqrt{n_1^2 - n_2^2}$$

Acceptance angle

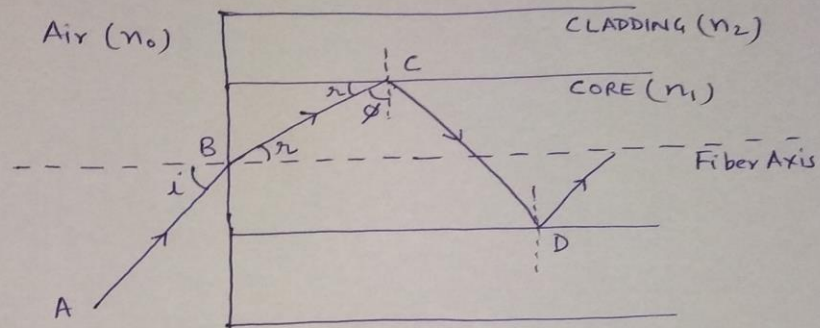


$\theta_{\max}$ is the maximum acceptance angle to the axis of the fiber at which light may enter into the fiber in order to propagate

Derivation of Acceptance angle and Numerical aperture:



Derivation of Numerical Aperture and Acceptance Angle



Applying Snell's law at point B,

$$n_0 \sin i = n_1 \sin r \quad \text{--- (1)}$$

We have 'at C', $\angle r + \angle \phi = 90^\circ$

$$\angle \phi = \frac{\pi}{2} - \angle r$$

We know that, for critical angle ' θ_c ' at point C

$$n_1 \sin \theta_c = n_2 \sin 90^\circ$$

$$\sin \theta_c = \frac{n_2}{n_1}$$

$$\theta_c = \sin^{-1} \left(\frac{n_2}{n_1} \right) \quad \text{--- (2)}$$

For Total Internal Reflection at point C,

$$\phi > \theta_c$$

$$\therefore \phi > \sin^{-1}\left(\frac{n_2}{n_1}\right)$$

$$\text{or, } \phi < \cos^{-1} \sqrt{1 - \left(\frac{n_2}{n_1}\right)^2}$$

$$\cos \phi < \sqrt{1 - \left(\frac{n_2}{n_1}\right)^2}$$

Now using eq. (1),

$$n_0 \sin i = n_1 \sin \phi$$

$$n_0 \sin i \leq n_1 \sin(90^\circ - \phi)$$

$$n_0 \sin i \leq n_1 \cos \phi$$

For maximum value of angle $i \Rightarrow i_m$
(Acceptance Angle)

$$n_0 \sin i_m = n_1 \cos \phi$$

$$= n_1 \sqrt{1 - \left(\frac{n_2}{n_1}\right)^2}$$

$$\sin i_m = \frac{n_1}{n_0} \sqrt{1 - \left(\frac{n_2}{n_1}\right)^2}$$

$$= \frac{n_1}{n_0} \sqrt{\frac{n_1^2 - n_2^2}{n_1^2}}$$

$$= \frac{n_1}{n_0 \cdot n_1} \sqrt{n_1^2 - n_2^2}$$

$$\boxed{\sin i_m = \frac{1}{n_0} \sqrt{n_1^2 - n_2^2}}$$

For air medium, $n_0 = 1$

$$\sin i_m = \sqrt{n_1^2 - n_2^2}$$

$$i_m = \sin^{-1}(\sqrt{n_1^2 - n_2^2})$$

where $i_m \rightarrow$ acceptance angle

$\sqrt{n_1^2 - n_2^2} \rightarrow$ Numerical Aperture

$$i_m = \sin^{-1}(N.A.)$$

Numerical Aperture is the property of optical fiber and it remains fixed irrespective of medium. However acceptance angle varies according to the medium where the optical fiber is kept.

Applications of fiber optic cables:

Some of the most popular uses of fibre optic cables are listed below.

1. As Internet cable:

Fibre optic cables transmit large amounts of data at very high speeds. As compared to traditional copper wires, fibre optic cables are less bulky, lighter, more flexible, and carry more data.

2. Computer Networking:

Networking between computers in a single building or across nearby structures is made easier and faster with the use of fiber optic cables. Users can see a marked decrease in the time it takes to transfer files and information across networks.

3. Surgery and Dentistry:

Optical communication is an important part of non-intrusive surgical methods, popularly known as endoscopy. In such applications, a minute, bright light is used to light up the surgery area within the body, making it possible to reduce the number and size of incisions made. Fibre optics are also used in microscopy.

4. Automotive Industry:

Fibre optic cables play an important role in the lighting and safety features of present-day automobiles. They are widely used in lighting, both in the interior and exterior of vehicles. Because of its ability to conserve space and provide superior lighting, fibre optics are used in more vehicles every day. Also, fibre optic cables can transmit signals between different parts of the vehicle at lightning speed. This makes them invaluable in the use of safety applications such as traction control and airbags.

5. Telephone:

Calling telephones within or outside the country has never been so easy. With the use of fibre optic communication, you can connect faster and have clear conversations without any lag on either side.

6. Lighting and Decorations:

The use of fibre optics in the area of decorative illumination has also grown over the years. Fibre optic cables provide an easy, economical and attractive solution to lighting projects . As a result, they are widely used in lighting decorations and illuminated Christmas trees.

7. Mechanical Inspections:

Fibre optic cables are widely used in the inspection of hard-to-reach places. Some such applications are on-site inspections for engineers and also inspection of pipes for plumbers.

8. Cable Television:

The use of fibre optic cables in the transmission of HD cable signals because they have greater bandwidth and speed. Also, fibre optic cables are cheaper as compared to the same quantity of copper wire.

9. Military and Space Applications:

With the high level of data security required in military and aerospace applications, fibre optic cables offer the ideal solution for data transmission in these areas.